

EXECUTIVE SUMMARY

Complete Business Analysis: Customer Churn, Property Pricing & Sales Performance

Project Overview

This project presents an end-to-end business analytics study covering three business areas: **customer churn, property pricing, and sales performance**. The objective was to transform business data into actionable insights using data cleaning, exploratory data analysis, statistical analysis, and machine learning techniques.

Business Objectives

- Identify factors associated with customer churn and develop a churn prediction model.
- Analyze property characteristics and develop a model for estimating property prices.
- Evaluate product and regional sales performance.
- Validate observed patterns using statistical techniques.
- Translate analytical findings into practical business recommendations and an implementation plan.

Data & Methodology

The project uses three datasets: **500 customer churn records, 300 property records, and 100 sales records**. The datasets were analyzed independently because they represent different business modules and do not share a common row-level structure.

The analysis included descriptive statistics, exploratory data analysis, correlation analysis, one-way ANOVA, logistic regression, and linear regression. Multiple professional visualizations were created to identify patterns and relationships in the data.

Key Findings

Customer Churn:

A logistic regression model achieved **93.0% accuracy, 75.0% precision, 54.5% recall, 63.2% F1-score, and 0.982 ROC-AUC** on the test set. The model correctly identified 6 of 11 actual churners, while 5 churners were missed. This indicates that the model can support churn-risk identification, while further model tuning may be useful when higher churn recall is required.

Property Pricing:

A linear regression model using area, bedrooms, bathrooms, age, location, and property type achieved an **R² of 0.941**, with **MAE of 2,188,736.34** and **RMSE of 2,907,633.21** on the test set. The results indicate that the selected property characteristics provide substantial predictive information for property prices in this dataset.

Sales Performance:

Total sales were **12,365,048** across **478 units**, with an average transaction value of **123,650.48**. Laptop generated the highest total sales at **3,889,210**, while North recorded the highest regional sales at **3,983,635**. Quantity and total sales showed a Pearson correlation of **0.688**, while price and total sales showed a correlation of **0.646**.

A one-way ANOVA comparing sales across East, North, South, and West produced an **F-statistic of 2.164** and **p-value of 0.097**. Since $p > 0.05$, there is insufficient statistical evidence to conclude that average sales differ significantly across the four regions.

Business Recommendations

- Develop an early-warning process using churn probabilities to support customer retention activities.
- Use property price predictions as a supporting reference when evaluating and pricing properties.

- Monitor product-level sales and product-region performance to support inventory and promotional planning.
- Track average transaction value, total sales, churn rate, and model performance as key business KPIs.
- Continuously validate models against new data and retrain them when business patterns change.

Expected Business Impact

The proposed analytical framework can support **data-driven customer retention, more consistent property pricing, improved sales monitoring, and continuous business performance tracking**. The results provide a foundation for further analysis using larger and more representative datasets.